LOCK-FREE SEQLOCK BOUNDARY CONTRACT & PREEMPTION RECOVERY

Asynchronous Inter-Core State Exchange with Zero Wait States and Deterministic Fallback Hierarchy

(a) Lock-Free Seqlock Publication Protocol

Atomic Barriers · Zero Mutex

UNTRUSTED WRITER (NPU/MPU)

1. **atomic_store_release(S, 2k+1)**
   Marks sequence counter ODD
2. **Store-Release Barrier**
   Flushes counter visibility
3. **Write Payload Fields**
   m_cmd, k, t_prod, Δt_valid, CRC
4. **Store-Release Barrier**
   Ensures payload commits first
5. **atomic_store_release(S, 2k+2)**
   Marks sequence counter EVEN

SAFETY READER (Real-Time MCU)

1. **S1 = atomic_load_acquire(S)**
   Sample initial sequence
2. **if (S1 & 1) reject();**
   Odd → Write in progress! Zero wait
3. **Load-Acquire & Copy**
   Copy payload to private stack
4. **S2 = atomic_load_acquire(S)**
   Sample second sequence
5. **if (S1 != S2) reject_torn();**
   Torn write trap (zero blocking)

SELF-CONTAINED BOUNDARY CONTRACT PAYLOAD (256 bytes)

`dot_m_cmd`	[float32]	Commanded mass flow target (physical limits [0, 45 kg/s])
`seq_id (k)`	[uint64]	Monotonic proposal index (detects dropped/skipped frames)
`t_prod`	[uint64]	Monotonic hardware timestamp of inference completion
`Δt_valid`	[uint32]	Lease expiration duration (e.g. 45.0 ms maximum validity)
`crc32_hash`	[uint32]	Payload integrity checksum (detects corrupted bitflips)

(b) Preemption Points & Deterministic Fallbacks

Guaranteed Bounded Response

WRITER PREEMPTION / CRASH POINT ANALYSIS

Point A · Crash Before Step 1 [S is even (2k)]
Reader sees intact prior record (2k==2k). Zero corruption. ✓

Point B · Preempted During Write [S is odd (2k+1)]
Reader tests (S1 & 1) != 0, drops buffer in <5 μs. Zero blocking. ✓

Point C · Crashed Mid-Publication [S stays odd]
Reader detects persistent odd sequence, trips fallback clamp. ✓

Point D · Write Fully Committed [S is even (2k+2)]
Reader validates S1==S2==2k+2 and CRC. Accepts fresh packet. ✓

4 BOUNDARY FAILURE CLASSES & DETERMINISTIC RESPONSES

1. MALFORMED: CRC mismatch / Out-of-bounds dot_m_cmd
Fallback Action: Clamp valve immediately to last verified safe orifice position.

2. STALE: t_now > t_prod + Δt_valid (Lease expired)
Fallback Action: Hold position for τ_hold = 10 ms; if persists, begin ramp-down.

3. MISSING: Sequence gap (k_new > k_prev + 1)
Fallback Action: Flag queue-overrun; apply state extrapolation with rate limit.

4. TORN / OVERRUN: Seqlock mismatch (S1 != S2)
Fallback Action: Reject torn read instantly (<5 μs); continue 1 kHz control tick.